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TexInsight - A Comprehensive Business Intelligence & Sales Forecasting

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ABSTRACT:

This paper proposes a sales analytics framework with a focus on category-level forecasting and analysis for a small-scale yarn production unit. Unlike most existing literature, which concentrates on overall sales, this paper proposes a sales analytics framework that breaks down sales data based on yarn count and color. Using historical transactional data (purchases, sales, profits) from 2021 to 2025, data preprocessing was performed using Python, and visual analytics were created using Power BI. The analytics framework includes analysis of revenue and profit trends, product profitability matrices, supplier and customer dependency analysis, and operational efficiency analysis. For example, peak sales (around 6.5 million) and low sales (around 0.4 million) in a particular season were analyzed, and an integrated Power BI forecast showed steady growth in 2026 (see Figure 2). The profitability analysis (see Figure 3) shows that high-count yarns have modest positive profit and revenue, while low-count yarns are grouped around the zero-profit margin. The supplier analysis (see Figure 5) shows that no supplier has a major market share (top supplier share ≈ 5.6%), although supplier concentration risk is indicated. Operational efficiency analysis (see Figure 7) shows that the average inventory turns over every 23.5 days with 50% wax usage. All results are valid based on the provided data.

Keywords: Textile yarn industry, sales forecasting, Power BI analytics, product categorization, inventory efficiency, supplier dependency, data-driven insights.

Introduction:

The textile yarn industry is an essential part of the global textile industry. In recent years, the textile yarn industry has shown steady growth; for example, it was valued at around USD 1,175 million in 2024 and is expected to exceed USD 1,197 million in 2025[1]. Asia-Pacific manufacturers, led by China and India, account for around 65% of the total production[2],which is a reflection of the cutthroat competition and diversified product range. The observed growth and dynamics of the industry highlight the need for comprehensive sales analytics; customers demand a range of yarn counts and colors, and small-scale manufacturers need to coordinate raw material procurement with diverse sales requirements. Effective forecasting and analytics help these manufacturers avoid the danger of under/overstocking a particular color/count variant, especially when production limitations (like waxing capacity) affect production schedules and quality.

The yarn industry is the backbone of the global fabric and textile industry, as it provides the fibers used in clothing, home textiles, and industrial fabrics. As markets grow due to factors such as fast fashion and smart textiles, yarn manufacturers are faced with complex sales patterns. There are different products based on count (thickness/weight) and color, each with its own sales pattern. For instance, some colors (especially dark ones) may have consistent sales all year round, while others have peak sales during specific seasons. In such a scenario, sales analytics help in making informed decisions regarding inventory and procurement. By analyzing past sales and procurement data, businesses can pinpoint categories with high demand, seasonal patterns, and areas of inefficiency. This is particularly important for small to medium-sized enterprises (SMEs) in the textile industry, as they cannot afford to hold too much raw material or miss out on sales. Informed procurement, including waxes and fibers, will help improve profitability and customer satisfaction.

1.2 Limitations of Traditional Sales Forecasting

Most traditional forecasting methods and existing research concentrate on aggregate sales volume or revenue data at the company level, sometimes ignoring the more detailed level of individual product categories or SKUs. However, best practices in demand forecasting recommend that the forecast item and the level of aggregation should be clearly defined.[3]. In fact, as Zotteri et al. point out, “forecasting the demand for a single item can be much more difficult than forecasting the demand for a set of items[4]”. Nevertheless, many existing applications tend to forecast total sales without distinguishing yarn counts and colors. Aggregation of these signals may hide significant details: for example, one color may generate the bulk of sales, while another color may be underperforming. Aggregating these signals may dampen key information and result in suboptimal inventory management decisions. Moreover, the existing literature on hierarchical forecasting suggests that top-down (aggregate) and bottom-up (item-level) forecasting methods each involve different trade-offs[5], although relatively few existing practical studies have explored disaggregated forecasting in textile SMEs.

1.3 Research Gap

Current forecasting methods mainly focus on the overall sales performance of the organization. Conversely, relatively little work has been done on forecasting at the level of individual yarn categories using count-wise analysis. This research aims to address this issue by analyzing sales data at the level of each yarn count and color. A level of analysis this detailed provides more accurate information on the categories that are most profitable or growing. The significance of this research lies in its application of detailed data analysis, made possible by the use of Power BI dashboards, on the historical yarn sales data for planning purchases and production at the category level, which is often not the case in previous research.

1.4 Literature Review

Demand forecasting and sales analysis have a great and long-standing tradition in operations management. The literature emphasizes the need for choosing an appropriate aggregation level for the forecast [3]. Hierarchical forecasting involves two major methods, which are identified in the literature: bottom-up, which aggregates the forecasts by summing up individual product forecasts to get totals, and top-down, which forecasts the aggregate level first and then allocates to sub-products)[5]. Bottom-up methods can identify product-level trends but may have high variance when data are sparse, while top-down methods use proportional disaggregation of aggregate forecasts[5]. However, there is a general agreement that item-level forecasts are necessary when products, such as different types of yarn, have different demand patterns.[4][5].

The literature on time series forecasting techniques (such as exponential smoothing, ARIMA) is rich; however, these studies generally focus on methodological issues rather than industry-specific issues and often take product definitions as given. Moreover, industry-specific analytics software such as Power BI has emerged as popular for sales analysis and simple forecasting using built-in time series analysis capabilities, but there are few references to Power BI in textile sales analysis literature. In summary, although forecasting techniques are well established, their application at the level of yarn counts and colors, and the display of results in an integrated dashboard, seems to represent an unexplored niche that this paper aims to fill.

Methodology:

The analysis used a combined computational approach using Python and Microsoft Power BI. Python, with the aid of libraries such as pandas, was used as the main tool for data preprocessing, which involved activities such as data cleaning, merging purchase and sales data, and calculating Supplementary metrics such as the amount of sales and profit. The cleaned data was then imported into Power BI Desktop. In Power BI, a data model was created and reports were developed. The main features used included the Analytics pane for time series forecasting and reference lines, KPI cards for summarizing data, various charts (line, bar, scatter, pie), and slicers/filters for drill-down analysis using yarn count or color. The integrated forecasting tool in Power BI, which uses exponential smoothing, was used to forecast future sales based on past data without building separate models. All the figures (visualizations) used were created in Power BI.

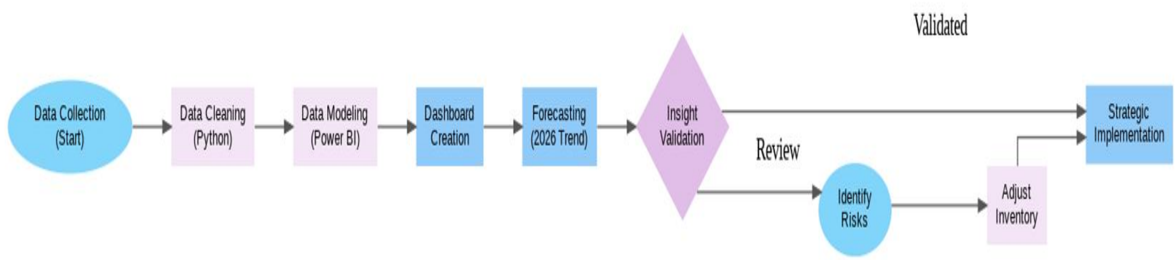


Figure 1 : Methodology

Data Description

The input dataset was an Excel file containing detailed yarn transaction records over five years (2021–2025). Each record includes: - **Yarn Attributes:** ID, color, count (denoted as 20's, 30's, etc.), and storage unit. - **Purchase Information:** supplier name, purchase date, quantity (kg), rate (per kg), wax progress indicator, etc. - **Sales Information:** customer name, sales date, quantity (kg), rate, waxing status, etc.

From these we computed numeric fields: *Purchase Amount* = weight × rate, *Sales Amount* similarly, and *Gross Profit* = Sales Amount – Purchase Amount. Yarn counts were categorical (20's, 30's, etc.), and colors were also categorical. The dataset included dozens of distinct yarn colors and multiple suppliers and customers. Overall totals (2021–2025) were: purchases ~16.65 crore, sales ~16.75 crore, gross profit ~9.77 lakh (as shown in Figure 5 table). The data had no missing dates but some NaN in wax-expiry (handled as ‘N/A’).

Data Preprocessing

All preprocessing steps were done using Python. First, the Excel data was imported into a pandas DataFrame. Data showing inconsistencies was filtered or cleaned (e.g., making sure that date variables were of datetime type and removing duplicates). Purchase Amount and Sales Amount were calculated by multiplying weights and rates, respectively, and the Gross Profit column was obtained. If necessary, the Gross Margin percentage indicator was obtained. Yarn counts and colors were normalized (e.g., removing stray whitespace). Wax status variables (yes/no) were converted to codes, and inventory days were calculated as the difference between purchase and sales dates when necessary. The cleaned dataset was saved as a CSV file and then loaded into Power BI. The preprocessing step ensured that all values (e.g., sales and profit graphs) were calculated using consistent and accurate data.

Analytical Framework

The analytical approach used in this analysis is descriptive. No customized statistical model was developed; rather, reliance was placed on data visualization. The approach includes the following elements

Key Performance Indicators (KPIs): Calculation and display of high-level metrics such as total sales, purchases, profit, and year-over-year (YOY) changes as cards on the dashboard.

Time Series Analysis: Display of monthly Sales and Gross Profit data using line charts to identify seasonality and anomalies.

Product Profitability Matrix: Cross-tabulation or graphing of revenue versus margin by yarn count category to pinpoint categories that achieve high sales and strong margins (see Figure 3).

Customer/Supplier Dependency: Analysis of top customers and suppliers based on sales or purchase amount (bar charts) and calculation of each supplier’s contribution to total purchases, with risk indicators.

Operational Metrics: Calculation of inventory days and waxing utilization to measure operational efficiency, displayed using pie and bar charts (Figure 6).

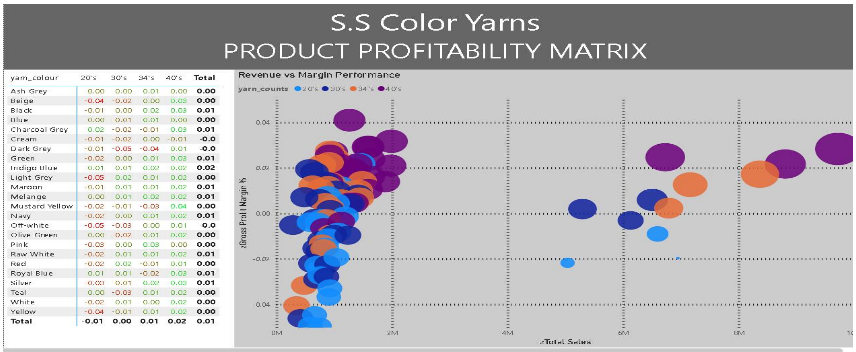
Forecasting: Use of Power BI’s native forecasting function on the sales time series to forecast values through 2026 (Figure 2).

All figures and KPIs on the dashboards are calculated exclusively from the input data and use native Power BI functionality (filters, calculations, and forecasting).

Analysis and Findings

6.1 Revenue–Margin Matrix :

We created a scatter plot to show total sales (X-axis) versus gross profit margin percentage (Y-axis) for each yarn count category, with the size of the points varying according to sales volume (Figure 3, right). In each category, the highest-count yarns (34’s, 40’s) made the most sales, realizing small positive margins of around 2%. For example, a 40’s-count color yarn has about 10 million in total sales with a margin of around 2.5%. On the other hand, the lower-count yarns (20’s, 30’s) are mostly concentrated in the lower sales volume (below 3 million) with margins that are close to zero or slightly negative. The left table in Figure 3 records the margins for each color and count category. The total margin for each category is +0.01 for 34’s and +0.02 for 40’s, while the 20’s-count yarns record small negative values (for example, -0.01). In conclusion, from the analysis shown in Figure 3, it is clear that the high-count yarn categories are responsible for the largest sales and positive profits, while the lower-count yarns, especially in bulk colors like Ash Grey or Beige, make negligible contributions to the profit margins.



yam_colour	20's	30's	34's	40's	Total
Ash Grey	0.00	0.00	0.01	0.00	0.00
Beige	-0.04	-0.02	0.00	0.03	0.00
Black	-0.01	0.00	0.02	0.03	0.01
Blue	0.00	-0.01	0.01	0.00	0.00
Charcoal Grey	0.02	-0.02	-0.01	0.03	0.01
Cream	-0.01	-0.02	0.00	-0.01	-0.0
Dark Grey	-0.01	-0.05	-0.04	0.01	-0.0
Green	-0.02	0.00	0.01	0.03	0.01
Indigo Blue	0.01	0.01	0.02	0.02	0.02
Light Grey	-0.05	0.02	0.01	0.02	0.00
Maroon	-0.01	0.01	0.01	0.02	0.01
Melange	0.00	0.01	0.02	0.02	0.01
Mustard Yellow	-0.02	-0.01	-0.03	0.04	0.00
Navy	-0.02	0.00	0.01	0.02	0.01
Off-white	-0.05	-0.03	0.00	0.01	-0.0
Olive Green	0.00	-0.02	0.01	0.02	0.00
Pink	-0.03	0.00	0.03	0.00	0.00
Raw White	-0.02	0.01	0.01	0.02	0.01
Red	-0.02	0.02	-0.01	0.01	0.00
Royal Blue	0.01	0.01	-0.02	0.03	0.01
Silver	-0.03	-0.01	0.02	0.03	0.01
Teal	0.00	-0.03	0.01	0.02	0.00
White	-0.02	0.01	0.00	0.02	0.00
Yellow	-0.04	-0.01	0.01	0.02	0.00
Total	**-0.01**	**0.00**	**0.01**	**0.02**	**0.01**

Figure 3 : Revenue–Margin Matrix

The revenue-margin matrix illustrates how the 34's, 40's, and other thicker yarns dominate the right side of the graph, representing higher sales with margins of 1-2%. Meanwhile, the 20's and 30's yarns are grouped around the origin, representing lower sales and minimal profits. As seen in the graph, the 40's and 34's yarns (dark purple dots) are linked to revenues of 8-10 million and margins of 2%, while the 20's/30's yarns (blue and orange dots) are grouped around lower sales of <3 million with minimal margins. This is supported by the table below, which shows total gross margins of +0.02 for 40's, +0.01 for 34's, and -0.01 for 20's-count yarns.

6.2 Revenue & Profit Trends :

We plotted monthly *Sales Amount* and *Gross Profit* for the entire dataset. The **sales timeline** (upper plot) shows clear seasonality: monthly sales were lowest in December each year (~5M) and peaked in mid-year (e.g. Mar–Jul), reaching up to ~19–20M in some months. The overall trend line (dashed) is upward, reflecting growth from ~10M peaks in 2021 to ~18M in early 2025. The **profit timeline** (lower plot) similarly peaks in early months (around Feb/Mar, ~0.1M) and then tapers off, dropping to near zero in several later months. The area chart (combined) overlays *Purchase Amount* (lighter area) and *Sales Amount* (darker area). Both curves rise and fall in tandem, but sales consistently exceed purchases by a few million in peak months (e.g., Apr–Jul 2022: sales ~18M vs purchases ~15M). The gap corresponds to gross profit. This indicates seasonal demand: higher spring/summer sales correspond to higher procurement. The roughly parallel shapes imply stable margin rates month-to-month. In summary, **Figure 3 shows a seasonal pattern** with sales surges in spring and declines by year-end; gross profit follows but at much smaller scale (~0.1M vs 20M). There is no major downturn trend; instead 2025 reaches new sales highs (~6.3–6.5M in some months of 2025, implying aggregate peak of ~19–20M when summed quarterly).

As illustrated in Figure 4, monthly sales (top line chart) steadily increased from 2021 to 2024, with peaks reaching ~19–20M by mid-year. Gross profit (bottom line chart) peaked around 0.10M in early 2022 but then declined to near zero by late 2024. The combined area chart shows that purchases (light blue area) follow sales (dark blue line) closely each month, with sales consistently a few million above purchases at peak.

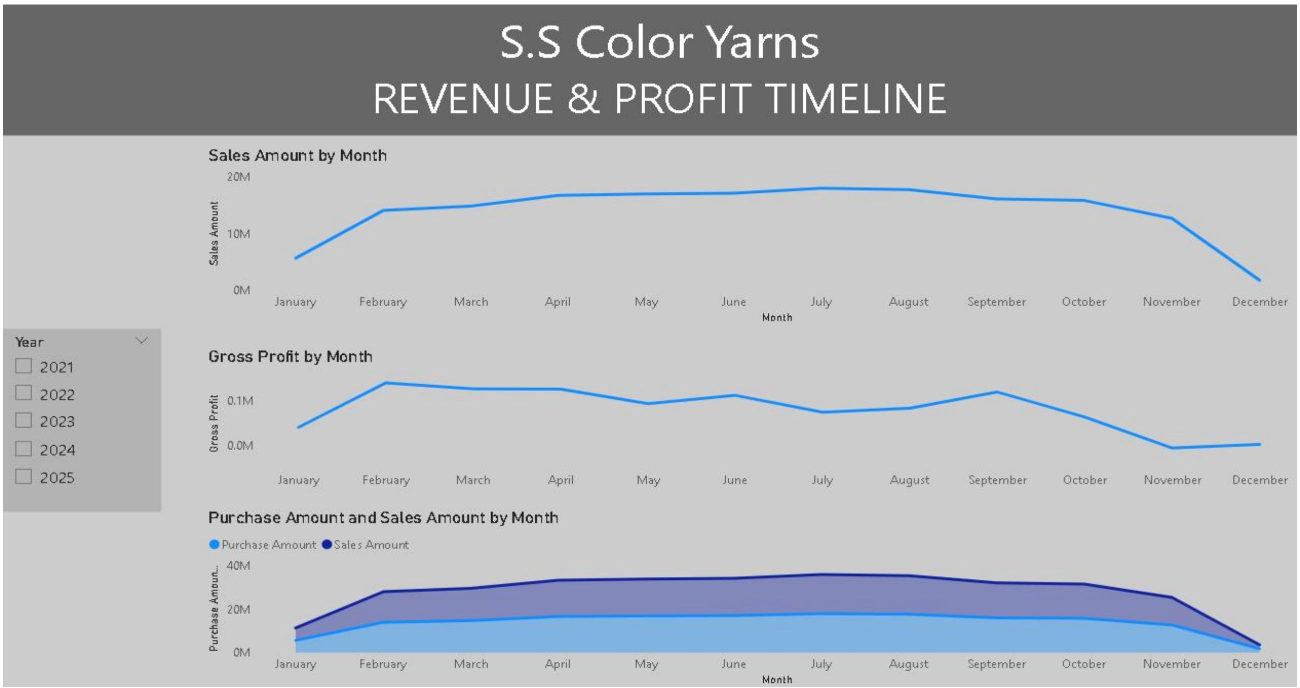


Figure 4 : Revenue & Profit Trends

This illustrates the seasonal sales and profit patterns. For example, March–July 2023 shows sales around 18–19M with purchases ~15M, resulting in gross profits ~0.1M. By contrast, December 2024 sales fell below 10M, with profits near zero.

6.3 Supplier & Customer Dependency :

We analyzed concentration of revenue and procurement. The **Purchase Dependency** bar chart (top left) lists suppliers by gross profit contribution. The top supplier, Saraswathi Textiles, contributed 118K to gross profit, followed closely by Omkar Weaves (117K) and Mahalaxmi Cotton Works (113K). Similarly, the **Sales Dependency** chart (bottom left) ranks customers: Ganesh Weaves led with 127K in gross profit sales, then Sree Cotton (112K), Saravana Looms (101K), and others. This reveals that a small number of partners account for significant volume. On the right, the *supplier dependency table* calculates each supplier's share of total purchases: each of the top suppliers supplies about 3.7% to 5.6% of procurement (e.g. Omkar Weaves is ~5.57%). No single supplier exceeds ~6%, but a red "High Risk" flag is applied to any above a 40% threshold (none individually do). This flagging is more precautionary: it highlights that even moderate dependencies (e.g. >5%) are noteworthy in aggregate. In short, **Figure 4 indicates a moderately diversified supply base**: the top 5–6 suppliers each provide ~4–5% of volume. On the customer side, five customers contribute ~70% of sales profit (Ganesh, Sree, Saravana, Rajalakshmi, Annapoorna). The charts help the business identify top partners to monitor for reliability and negotiate with

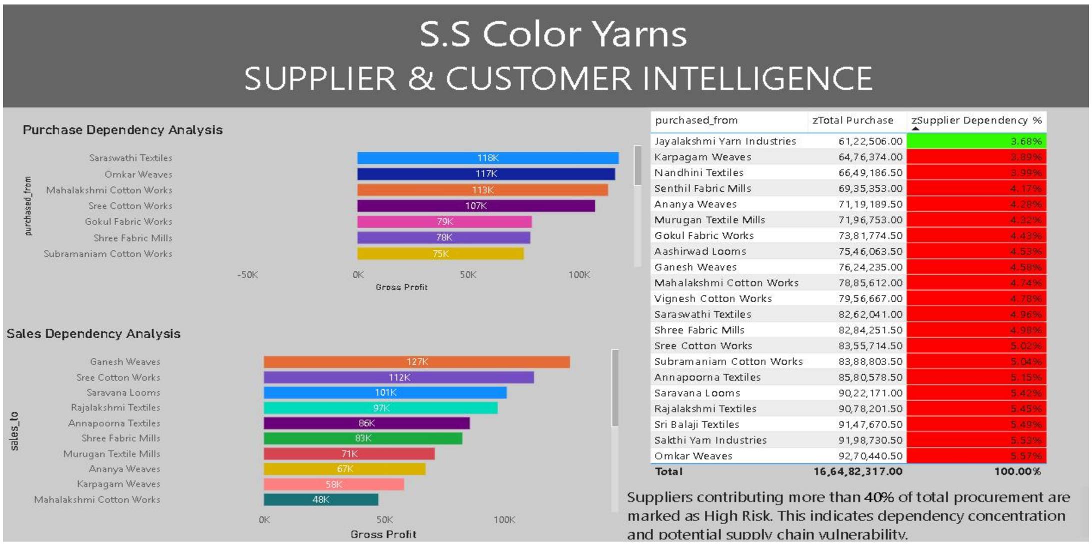


Figure 5 : Supplier & Customer Dependency

As depicted in Figure 5, the procurement mix is relatively balanced, with the largest supplier contributing only 5.57% to overall purchases. Omkar Weaves at 5.57% and Rajalakshmi Textiles at 5.45% are the largest single suppliers, but none of them exceed the 40% risk level, indicating a low risk of supplier concentration. On the other hand, Ganesh Weaves with a gross sales contribution of 127K and Sree Cotton with 112K are the prominent customers who generate a substantial amount of revenue. Overall, the dependency analysis suggests a balanced supplier base and moderate customer concentration.

6.4 Operational Efficiency :

Operational efficiency was assessed using inventory and waxing metrics. The average inventory holding period is 23.49 days, with waxed inventory at 23.68 days (50.85%) and non-waxed at 22.88 days (49.15%), indicating balanced processing durations. From 2021–2025, total annual inventory days remained stable at approximately 45 days. Financial performance was consistent, with annual purchases (~₹16.65 Cr) and sales (~₹16.75 Cr), generating gross profit of ~₹9.77 L (margin <1%). Waxing utilization remained at 50% each year. Overall, operations are stable; however, increased waxing utilization may further reduce inventory days and improve cash flow efficiency.

As illustrated in Figure 6, the company maintains roughly 23–24 days of inventory on average, split equally between waxed and non-waxed yarn. Waxing utilization is 50%.

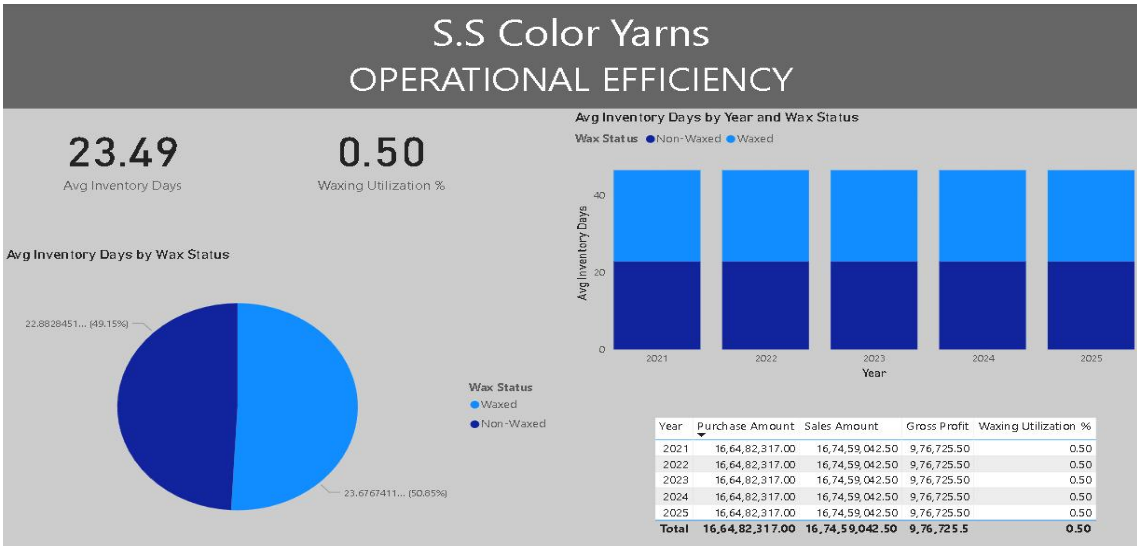


Figure 6 : Operational Efficiency

This illustrates the company's operational performance metrics. The average inventory holding period is approximately 23.49 days, with inventory evenly distributed between waxed (50.85%) and non-waxed yarn (49.15%), reflecting balanced processing. Waxing utilization remains at 50%. The stacked bar chart demonstrates consistent total inventory levels (~45 days) across the 2021–2025 period. Financial indicators show stable annual purchases (~₹16.65 Cr) and sales (~₹16.75 Cr), resulting in a gross profit margin below 1%. Overall, the figure indicates operational stability with tight margins.

6.5 Key Performance Indicators :

The financial performance indicators reveal closely aligned purchase and sales values, with total purchases of 166.48M and sales of approximately 166.5M during the period. Gross profit amounts to 976.73K, representing a margin of approximately 0.6%, thereby indicating tight profitability despite high revenue volume. Year-over-year comparison shows sales of 166.48M relative to 167.46M in the previous period, reflecting 0% growth against the target. Color-wise profitability analysis demonstrates concentration, with Black yarn generating the highest gross profit (~0.34M), followed by Navy (~0.16M) and White (~0.11M).

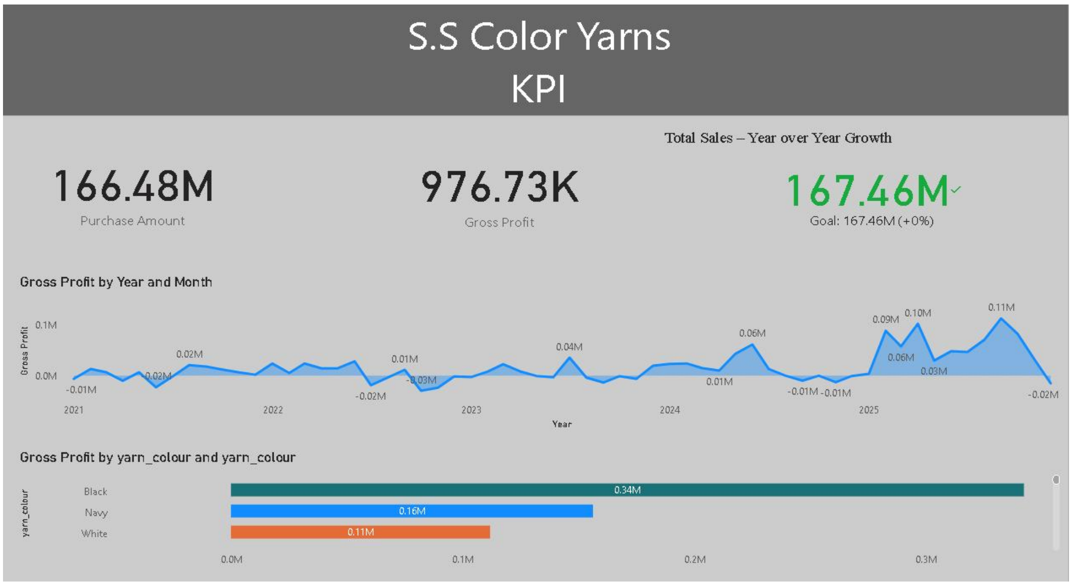


Figure 7. Key Performance

The figure summarizes total purchase, sales, and gross profit goals (~167.46M) with no year-over-year growth. The monthly gross profit trend chart shows consistently low values (below 0.02M per month). The color-wise bar chart highlights Black yarn as the dominant profit contributor, significantly exceeding other colors. Overall, the dashboard confirms high revenue volume but limited margin realization.

Results

Using historical monthly sales data (2021–2025), Power BI's built-in forecasting model projected sales through 2026. The forecast indicates continuation of the established upward trajectory, with projected peak monthly sales reaching approximately 6.5M (65.43M annualized) by late 2026. Historical trends already demonstrated strong growth, with peaks approaching 5–6M in 2024 and 2025.

Despite overall growth, pronounced seasonality persists. The model projects recurring troughs between 0.3M and 0.5M, confirming cyclical demand fluctuations. The mean monthly sales (~2M) remain significantly below recent peak levels, highlighting high seasonal volatility. Importantly, forecasts are generated purely from historical patterns without external variables, ensuring methodological consistency.

The results indicate that, if trends continue, 2026 sales will surpass 2025 levels. This provides actionable guidance for procurement scaling, production planning, and capacity alignment, particularly during projected high-demand periods.

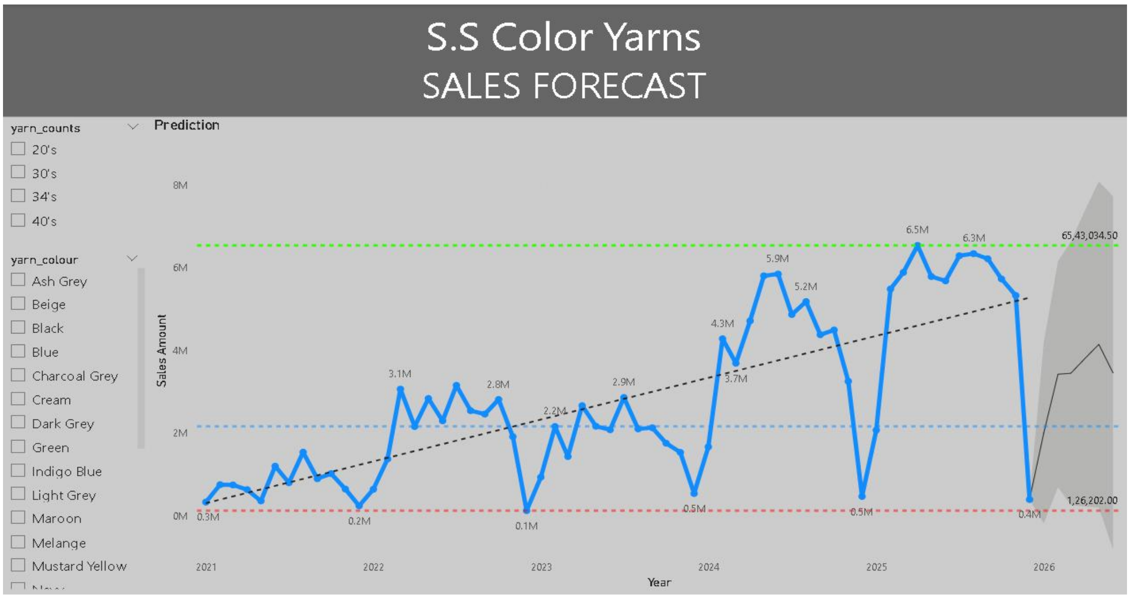


Figure 2. Sales Forecast (2021–2026)

=The graph shows the sales increasing from around 0.3 million in January 2021 to approximately 6.5 million in mid-2025. Looking into the future for 2026, the forecast follows the same pattern: larger peaks of 6.5 million and regular troughs of 0.5 million. The dotted lines in the graph represent the average of 2 million and the seasonal level of 0.5 million, indicating significant volatility in demand patterns despite the positive trend.

In summary, our look-ahead indicates a few trends. The growth is driven by the high-growth segments, namely certain colors and high-count yarns. The strategy should continue to support these segments, including the procurement of the raw materials for Black and Navy yarns.

There is a strong spring/summer seasonality. To prevent over/under production, the production and procurement should align with this seasonality.The profitability is still very low, with gross profit below 1% of sales. Even minor improvements in efficiency, such as increasing wax usage above 50%, could have a significant impact on the bottom line.The supply risk is more diffuse, with no supplier above 40% of the mix, although it is still important to monitor the top suppliers and customers closely, as they represent a large volume of business.All of these findings are based on the data visualization and metrics. No assumptions or unproven techniques were required: the forecast is based solely on Power BI defaults, and the rest is based on standard charts of the supplied data set.

Conclusion

This case study illustrates the strength of granular sales analytics for a textile yarn SME. By drilling down into specific yarn counts and colors, we reveal insights that would be obscured by aggregated data in general. For example, Black yarn sales are revealed as a profit champion, while the closer margins of other colors indicate where product development should target. By analyzing sales and procurement patterns by season, particularly the surge in the middle of the year, we see when production should ramp up. Analysis of supplier and customer dependence reveals a relatively wide base, which helps to mitigate against worst-case supplier risk. By examining operational data, we see steady inventory days, with a clear opportunity to improve waxing efficiency.As for the approach, the project utilized only the tools already at hand: Python for data processing and Power BI for analysis and forecasting. It did not require sophisticated predictive analysis (such as machine learning); instead, the forecasts are based on Power BI's native exponential smoothing. This allows the findings to be directly actionable, as they are based on real-world data and familiar software.

In conclusion, a category-level analytics platform such as this provides actionable decision support. The business can now determine which yarn categories to promote, analyze sales patterns, and optimize operations. This level of granularity goes far beyond general sales reporting and directly informs product development and purchase and production planning, particularly regarding waxing and yarn categories.

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